Algebraic Number Theory NMAL431
Homework 2

Due date: 08 December 2026 at 17:20

Please send any comments and/or corrections to `robin.visser@matfyz.cuni.cz`

1. Let p be a prime and $n \geq 1$ an integer. Let $\pi = \sqrt[n]{p}$ and let $A = \mathbb{Z}_p[\pi]$.

   (a) Prove that A is a complete discrete valuation ring, with uniformizer π and residue field $\mathbb{F}_p$.

   (b) Let $x \in A$. Recall from lectures that x admits a unique expression $x = \sum_{k=0}^{\infty} a_k \pi^k$ with $a_k \in \{0, 1, \ldots, p-1\}$ for each $k \geq 0$. Show that $x \in \mathbb{Q}(\pi)$ if and only if the sequence (a_k) is eventually periodic.

2. Calculate (using Hensel's lemma or otherwise) the number of roots of each polynomial in the corresponding field:

   (a) $X^3 + 1$ in $\mathbb{Q}_7$,

   (b) $X^2 + 2X + 4$ in $\mathbb{Q}_2$,

   (c) $3X^3 + X + 3$ in $\mathbb{Q}_3$.

3. Let D_1, D_2 be distinct squarefree integers, both $\neq 0, 1$, let $D_3 = D_1 D_2 / \gcd(D_1, D_2)^2$, and let $E = \mathbb{Q}(\sqrt{D_1}, \sqrt{D_2})$. E has Galois group $(\mathbb{Z}/2\mathbb{Z})^2$ and its three quadratic subfields are $\mathbb{Q}(\sqrt{D_1})$, $\mathbb{Q}(\sqrt{D_2})$, and $\mathbb{Q}(\sqrt{D_3})$.

   For each rational prime p, let e_p and f_p denote the ramification index and residue degree of a prime of $\mathcal{O}_E$ lying above p, and let g_p denote the number of primes of $\mathcal{O}_E$ lying above p.

   (a) Let p be an odd prime such that $p \nmid D_1 D_2$. Show that

   $$(e_p, f_p, g_p) = \begin{cases} (1,1,4) & \text{if } \left(\frac{D_1}{p}\right) = \left(\frac{D_2}{p}\right) = 1, \\ (1,2,2) & \text{otherwise.} \end{cases}$$

   (b) Let p be an odd prime such that $p \mid D_1 D_2$. Show that p divides exactly two of D_1, D_2, D_3, and let d be the third. Show that

   $$(e_p, f_p, g_p) = \begin{cases} (2,1,2) & \text{if } \left(\frac{d}{p}\right) = 1, \\ (2,2,1) & \text{if } \left(\frac{d}{p}\right) = -1. \end{cases}$$

   (c) Let $p = 2$. Show that 2 is unramified in E if and only if $D_1 \equiv D_2 \equiv 1 \pmod 4$, and that in this case 2 splits completely if and only if $D_1 \equiv D_2 \equiv 1 \pmod 8$ and has $(e_p, f_p, g_p) = (1, 2, 2)$ otherwise.

   (d) Find a number field K such that, for every rational prime p, there are an even number of prime ideals $\mathfrak{p}$ in $\mathcal{O}_K$ lying above p .